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Bistable Mechanisms 3D Printing for Mechanically Programmable Vibration Control

Ali Zolfagharian, Frédéric Demoly, Mohammad Lakhi, Bernard Rolfe, and Mahdi Bodaghi*

Bistable mechanisms, characterized by their ability to maintain two distinct stable states under external stimuli, offer a versatile solution to engineering challenges, particularly in vibration suppression systems. While tuned mass dampers (TMDs) effectively mitigate vibrations at specific frequencies, their efficacy falls short across a broader spectrum. This work proposes enhanced adaptability for passive vibration reduction without dependence on complex control systems or external power sources by integrating bistable mechanisms into TMD systems. A bistable mechanism is designed based on numerical simulations and is 3D-printed via fused filament fabrication. The bistable mechanism provides more adaptive characteristics by displacing the tuned mass on a flexible link. Following experimental testing, the optimized design for bistable mechanisms has been determined. Then, a chain of bistable mechanisms is introduced to achieve the desired displacement for carrying the TMD payload. The stroke length for the TMD vibration test has been customized to reduce system vibrations by carefully optimizing geometric configurations of bistable mechanisms.

1. Introduction

Mechanical vibration is a recurrent phenomenon in various engineering systems. It is defined as the oscillatory motion of structures out of their equilibrium position, caused by external forces, leading to restrictions on performance, potential structural damage, and even compromising the safety of the operator working with the system. Tuned mass damper (TMD) is a practical passive engineering solution designed to neutralize the vibrations and improve the performance of the operating system subjected to an external force.[1] There have been a few studies exploring the use of TMDs as passive vibration control devices for structural vibrations. TMDs are effective in mitigating oscillations induced by dynamic loads,[2,3] but their fixed mechanical properties limit their adaptability and suitability to changing dynamic conditions. A traditional TMD uses an additional mass–spring system connected to the main structure to dissipate the unwanted oscillation at a specific frequency. However, a major drawback of this solution is its limited effectiveness, as it operates only at the specific frequency for which it is designed. Traditional TMDs lose their efficiency when they encounter a wider range of frequencies. Altering parameters such as the weight or spring stiffness aids in adapting the TMD system to various oscillating frequencies.

The exploration of bistable designs unveils their potential for creating more responsive and efficient TMDs capable of dynamically adjusting to different vibrational frequencies.[4] This adaptability stems from the inherent characteristics of bistable mechanisms, where the transition between their two stable states can be harnessed to reposition the mass within the TMD system, thus altering its dynamic response. In response to this, researchers studied the integration of magnetorheological (MR) dampers with TMD systems for active vibration control.[5] MR dampers, known for their controllable damping properties, can dynamically adjust damping forces in response to real-time structural vibrations. This integration enables the development of MR-TMD systems capable of actively modulating damping characteristics based on instantaneous feedback from sensors monitoring structural dynamics. The MR-TMD system works better than traditional passive TMDs, reducing vibrations more effectively and being able to adapt to changing dynamic conditions. This innovative approach offers a robust solution for mitigating structural

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vibrations and enhancing system performance in various engineering domains.

Damanpack et al.[6] explored the stability of hyperelastic curved beams through a combination of experimental testing and numerical simulations. They fabricated curved beam specimens with varying opening angles using 3D printing techniques and subjected them to mechanical testing to characterize their behavior under different loading conditions. Finite element analysis (FEA) was used to predict the nonlinear behavior of structures. Results from experimental techniques were compared to verify the behavior of the structures as per the curvature beam. Further analysis of the performance prediction of the bistable mechanisms provided more information on their design.[7,8] The curved beam, a crucial component in compliance beam design, plays a significant role in exhibiting bistability performance.[9] The geometry and materials used in the design and fabrication process determine the component's capacity to store mechanical energy and exhibit bistability.[10,11]

The literature studies have shown that the choice of appropriate material directly impacts the component's performance[12,13] as it directly affects the bistability performance.[14,15] Ideally, a material with a high strength-to-modulus ratio, which is an important characteristic for compliant mechanisms, is desired because it allows a larger deflection before failure. Zirbel et al. have used acrylonitrile butadiene styrene (ABS) to create a prototype of a bistable mechanism, as it can demonstrate similar effects to other materials such as metallic glass or titanium, as well as the ease of fabrication, making it a good option for prototyping purposes.[16] Therefore, ABS is used as the material for the bistable component in this study.

Bistable mechanisms have been widely utilized in vibration control, particularly in the context of quasi-zero stiffness (QZS) for vibration isolation. Earlier studies[17–19] mainly explore metastructures with QZS properties by employing bistable unit cells with negative stiffness properties generated through bistable beams and curved geometries. These setups effectively reduce transmissibility, making them highly efficient for isolation applications where the goal is to minimize external vibration transfer to subject structures. While the focus of these studies is on QZS concepts for isolation, the present work uniquely applies a bistable mechanism within a TMD system for vibration suppression rather than isolation. Here, the bistable mechanism is designed to adaptively reposition the tuned mass along a flexible structure, leveraging the stroke of the bistable element to passively tune the TMD's frequency response. By dynamically shifting the location of the tuned mass, this work uses the bistability principle not from the payload viewpoint but to achieve adaptive vibration damping across varying frequencies, thereby enhancing the TMD's effectiveness in vibration suppression.

Using 3D printing technology to fabricate bistable components further amplifies the feasibility and innovation of this approach, offering a pathway to tuned vibration control solutions.[20] Reconfigurable structures using bistability properties were printed to serve as mechanical switches, simplifying actuation and motion control without the need for electronics-controlled systems.[21] Moreover, other researchers developed bistable components using shape-memory polymers, which allow twisting and rotational motion based on compliant mechanisms with built-in joints.[22,23] Bistable structures have two stable states separated by an energy barrier, known as snap-through instability. The transition between these stable states can be done by applying enough energy that passes the hill of the barrier, resulting in a stable state with lower energy without additional energy. The shape of the component remains in a stable position over time without any energy consumption.

However, existing studies primarily focus on bistable structures in static or quasistatic conditions and have not fully explored their potential applications for adaptive vibration control in dynamically loaded systems. Additionally, the existing literature lacks studies on using bistable mechanisms specifically within TMD systems to enhance adaptability across varying vibrational frequencies. By introducing bistable mechanisms into the TMD design, the system could achieve adaptive vibration suppression capabilities, thus broadening its applicability in fields where reliable and low-maintenance vibration control is essential, such as in aerospace, civil engineering, and automotive industries. Furthermore, leveraging 3D printing to create bistable components enhances the flexibility and feasibility of this approach, as it enables rapid prototyping and fine-tuning of designs.

In this work, a novel approach is proposed where the placement of the mass from the reference point determines the stiffness of the spring in that region. To change the position of the mass along the structure without adding any complex control system or electronic devices requiring an external power supply, bistable mechanisms are considered. A bistable mechanism is a mechanical system with two stable equilibrium positions that allows it to hold and switch between two distinct states based on an external stimulus. The distance between the two states is manipulated to move the TMD mass accordingly. The main objective of this research is to design and optimize a bistable component within a TMD system for a flexible stainless steel structure, therefore enhancing its attenuation performance under various oscillating frequencies without using any active control system. The work will report the study of multiple bistable designs, and related prototypes will be 3D printed. The efficient vibration suppression will be achieved through numerical simulations and optimization, then contributing to the advancement of vibration control technologies.

2. Methodology

2.1. Design of Bistable Mechanism (Cell)

Integrating cells and stacking bistable mechanisms in the article are justified as innovative strategies to enhance structural complexity and functionality. This approach allows for the creation of more adaptable and efficient systems with tailored mechanical responses, suitable for a wide range of applications that are used in this work. Designing gradient bistable cells based on prescribed maximum instability forces is a strategic approach to creating programmable multistable mechanical metamaterials.[24,25] Researchers have varied the thickness, geometry, or material properties of curved shells to develop these innovative materials, enabling the design of structures with complex, programmable behaviors suitable for a wide range of applications.[26] This method leverages the intrinsic properties of materials and geometric configurations to achieve the desired mechanical

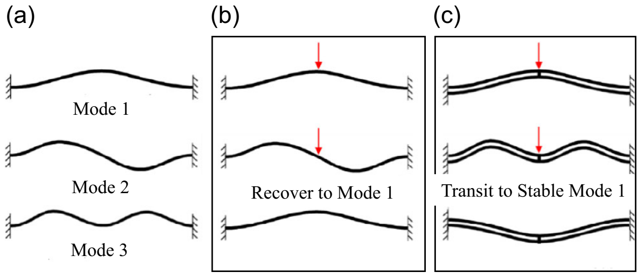

Figure 1. a) Different buckling modes of the compliance beam. b) Deformation of a single curved beam with recoverability. c) Deformation of a bistable double curved beam. Reproduced from[17] with permission of Springer Nature.

responses. The novelty of our approach lies in its precise control over the mechanical behavior of structures, allowing for the creation of programmable multistability.

Two conditions must be met for the curved beam to exhibit bistability,[27] i.e., the ratio of beam apex height-to-thickness must exceed 2.31 and the buckling mode 2 (**Figure 1**a) must be constrained.[17,28] Therefore, a double-curved beam system with a membrane connecting beams at the midspan was used in this work to ensure buckling mode 3 (Figure 1c) and bistability. The membrane transfers the rotational motion at either beam to the axial motion at the other beam. The motion patterns of the two beams are identical, resulting in reduced rotation. This reduces the generation of asymmetric modes and enables the suppression of buckling mode 2. Since buckling can take either an up or down direction, this creates two stable positions for the bistable mechanism.

Qiu et al.[27] proposed the profile of the curved beam in the shape of buckling mode 1 of a fixed-end elastic beam, with the configuration of the beam expressed as

$$l(x) = \frac{h}{2}\left[1 - \cos\left(2\pi\frac{x}{l}\right)\right] \tag{1}$$

where h is the apex of the cosine curve and l is the horizontal span of the curved beam. Besides the parameters that effect the profile of the curve beam, other parameters are also introduced to the formation of the component, including the beam thickness t and the distance gap between two beams g (**Figure 2**b). The straight beam in the middle and the side walls were expected to constrain the lateral movement of the curved beams.

A 3D model of the beam profile is generated using Autodesk Inventor (San Francisco, USA) with its parameters to comply with the profile configuration and constraints to maximize the ability to achieve bistability (Figure 1a). While there are constraints to follow during designing the compliance beam, there is no specific shape or outline that has been tested or proven to achieve bistability. Therefore, the shape of the compliance beam of the component is designed using simple curves. The beam is constructed with two connecting parts: the hinges that connect the walls to the beam and the curve beam that initiates the snap-through behavior (Figure 2).

The hinges are created using the tangent function with a 15 degree sweep path length; the circle radius starts at 20 mm and increments by 0.6 mm to create the thickness of a single beam;

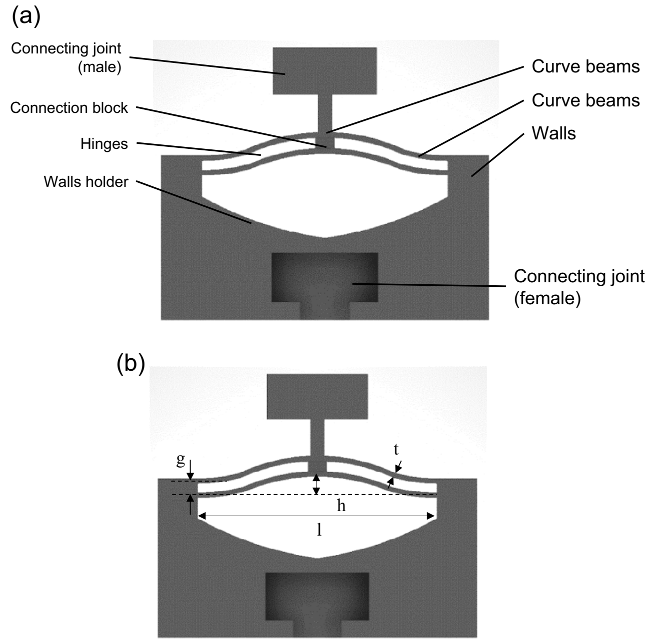

Figure 2. Schematic of the bistable component: a) 2D CAD drawing with annotations and b) 2D CAD drawing with dimensions.

each beam is 1.2 mm apart. The curved beam is designed using a 3-point arc with two of its ends connected to the hinges; the center of the curved beam is placed on the center line of the component with a radius of 23 mm. Once both beams are formed, a middle linking block between the beams is created to link the beam movements together, improving the response of the compliance beam.

2.2. Design of Stacking Mechanism

To improve the performance of the bistable component by varying the stroke length (the distance covered between two stable states), a stackable design is implemented that enables each unit to be connected by simply sliding the connecting joint together. The female joint is made with a 3% tolerance, ensuring the bistable cells can be connected firmly and will not loosen during vibration tests. A slot is also designed to fit the component onto a flexible link (**Figure 3**).

2.3. Equivalent Spring–Mass System of the Flexible Link TMD

The flexible stainless steel structure is modeled to represent the simple mass–spring mechanical system, and the TMD is implemented, as shown in **Figure 4**.[3] A simulation model of a mass–spring system with a TMD attached to the primary mass is created to observe the behavior of the main system and the performance of the TMD.

The primary system profile is set to be weight of the main mass $m_{\text{main}} = 0.15$ kg; spring constant $k_{\text{main}} = 226.8\ \text{N m}^{-1}$, therefore, the oscillating frequency of the main mass can be computed:

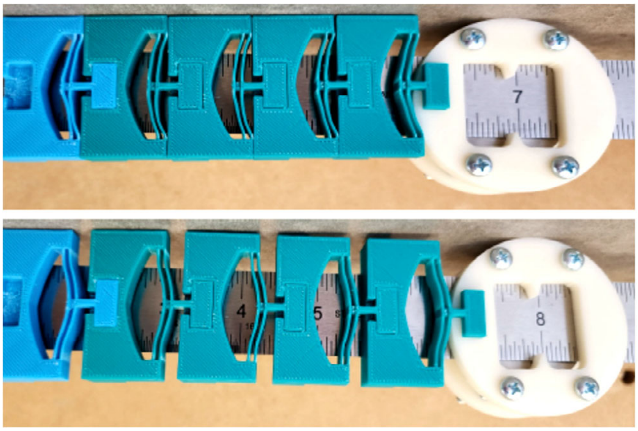

Figure 3. The stacking bistable mechanism in bistable positions.

$$\omega_{\text{main}} = \sqrt{\frac{k_{\text{main}}}{m_{\text{main}}}} \tag{2}$$

For the TMD system to successfully stabilize the main system, ideally, the natural frequency of the TMD ω_{TMD} should approximately equal the natural frequency of the main system. With that in mind, we can choose the TMD system profile such that its natural frequency is equal to ω_{main}. The calculated mass is 0.01875 kg. Using the natural frequency formula, we can get the value for the TMD spring constant using

$$k_{\text{TMD}} = \omega_{\text{main}}{}^2 \times m_{\text{TMD}} \tag{3}$$

When the TMD system is attached to the main mass, the TMD mass oscillates with the same natural frequency, which reduces the main mass's vibration; hence, the position of the main mass is significantly reduced. To reduce the vibration for a flexible structure, the parameters from the mass–spring system must be related to the flexible structure, as the stiffness of the spring is assumed to be proportional to its length. In the case of a flexible structure, the placement of the mass on the structure from the reference point determines the stiffness of the structure in that region; this relationship is described as the following equation:

$$k = \frac{3EI}{L^3} \tag{4}$$

where E is the constant modulus of elasticity of the material, L is the placement of the mass from the reference point, and I is the moment of inertia. For the numerical simulation and experiment, the flexible structure is represented by a 600 mm stainless steel ruler, so some parameters of the spring constant are given based on the material and the profile of the ruler, specifically, $E = 210$ GPa, where the base width of the shape is given as 0.02 mm and the height is given as 0.0015 mm with a rectangular cross section.

Given m_{main}, k_{main}, and m_{TMD}, the distance between m_{TMD} and m_{main} on the flexible stainless steel structure can be computed using the natural frequency formula $\omega_{\text{main}} = \omega_{\text{TMD}}$ and Equation (2) to come up with the relationship between the mass

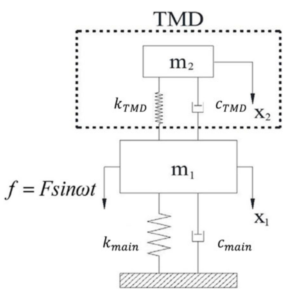

Figure 4. Proposed TMD system model.

Table 1. FFF 3D printing processing parameters.

Name	Value
Nozzle size [mm]	0.4
Filament size [mm]	1.75
Extruder temperature [°C]	240
Platform temperature [°C]	105
Infill percentage [%]	50
Infill pattern	Hexagon

Table 2. Experimental and simulation results comparisons.

Bistable design nos.	Results	Peak force in state 1	Peak force in state 2	Maximum force at end of test
#1	Simulation	7.05	5.89	42.58
	Experiment	6.07	5.41	42.79
	Error [%]	13.87	8.08	0.50
#2	Simulation	7.27	2.33	29.69
	Experiment	8.26	2.65	26.14
	Error [%]	13.64	13.87	11.95
#3	Simulation	28.32	13.27	61.95
	Experiment	32.30	15.07	61.04
	Error [%]	14.05	13.62	1.46
#4	Simulation	22.11	12.04	69.23
	Experiment	24.01	10.53	66.80
	Error [%]	8.06	12.45	3.50
#5	Simulation	22.21	17.06	31.14
	Experiment	24.94	18.48	35.40
	Error [%]	12.30	8.35	13.70
#6	Simulation	12.56	−1.90	25.08
	Experiment	13.06	−1.59	27.63
	Error [%]	3.98	15.87	10.17
#7	Simulation	18.36	20.98	97.30
	Experiment	20.72	21.97	104.35
	Error [%]	12.88	4.71	7.24
#8	Simulation	8.40	−6.28	103.01
	Experiment	8.96	−6.00	107.10
	Error [%]	6.78	4.44	3.97
#9	Simulation	12.55	−5.08	140.65
	Experiment	11.10	−5.36	118.17
	Error [%]	11.52	5.59	15.98

and spring stiffness of the main system for the best performance of vibration control.

3. Results and Discussions

During the fused filament fabrication (FFF) process, there are several parameters adjusted so the component can fit precisely on the scale of the vibration test and increase the ability to achieve bistability, easing the postprocessing steps of removing the support structures and excess materials, separating the model from the printing bed, etc. The main parameter that directly affects the performance of the component is the initial infill percentage, as it is also a factor that affects the rigidity of the walls. By increasing the infill percentage to 50% (with a hexagon infill pattern), the walls can be stronger and can withstand the force applied from

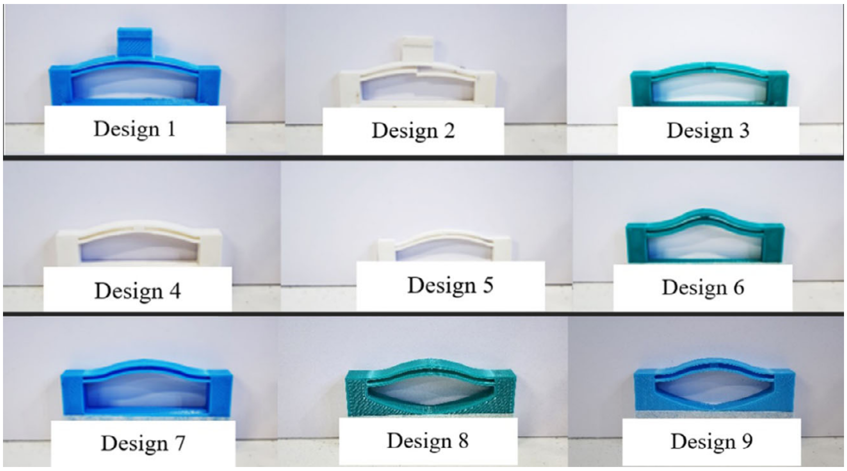

Figure 5. Different 3D-printed bistable designs.

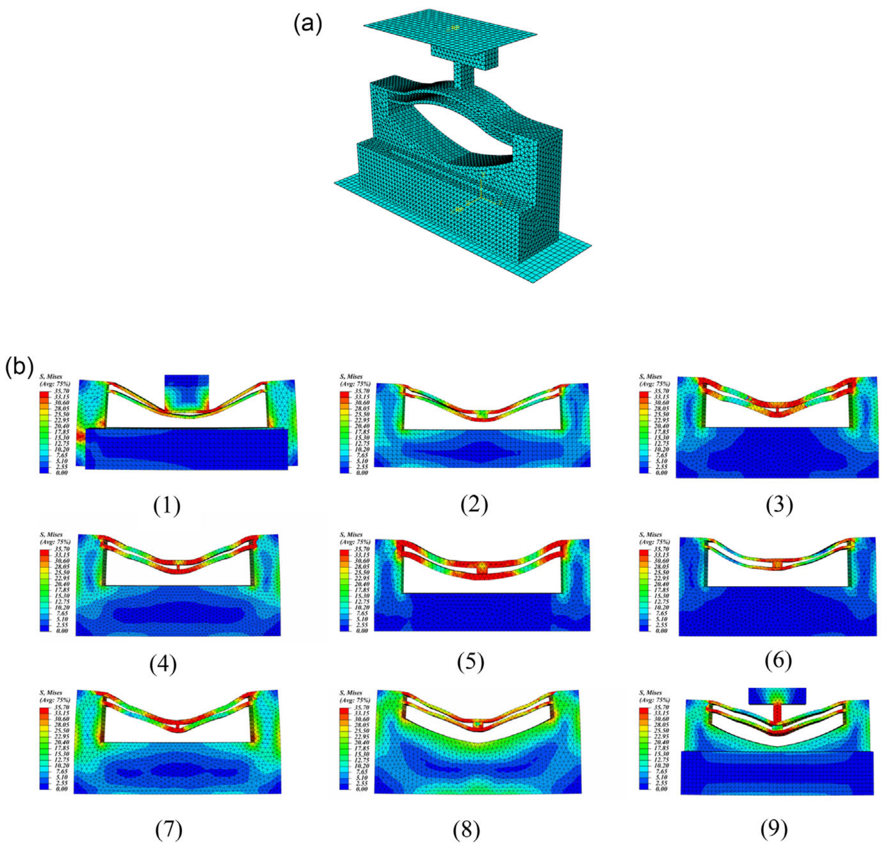

Figure 6. a) Meshing of a bistable mechanism sample. b) FEA results of the 3D-printed bistable designs.

the compliance beam during compression. The 3D printer used is the Flashforge Guider IIs (Jinhua, China), using a general 1.75 mm ABS filament, giving the component a suitable strength and flexibility for the fabrication of compliance components. The settings to use for the given printer and filament are shown in **Table 1**.

Multiple bistable components have been fabricated to test the functionality of the compliance beam. Each model is adjusted with different factors such as hinge angle, hinge length, curve profile, and printing infill, which influence the performance of the beam, all of which have been shown in **Table 2** and **Figure 5**.

Through the printing tests, each model is analyzed in terms of bistability, scale factor, and reliability of the beam. With each factor adjustment that is tested to have a positive effect on the component, it will be kept for the next redesign stage. Any factor that causes failure will be adjusted or eliminated in the next test to improve the quality of the bistable component.

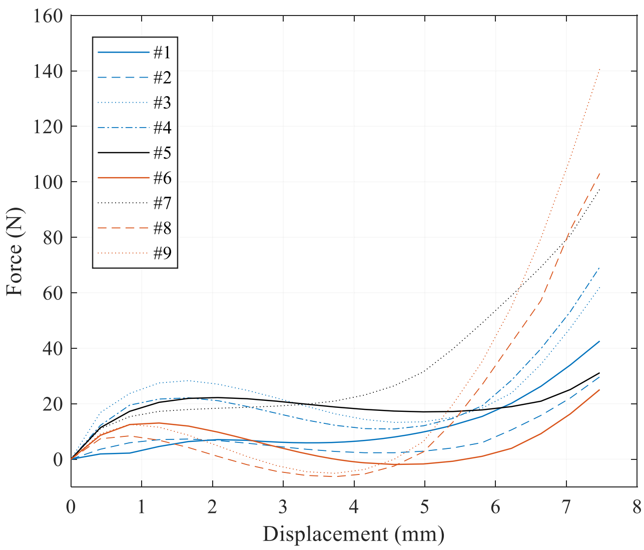

Figure 7. Force–displacement FEA results of the 3D-printed bistable samples.

Multiple experiments demonstrate the instability of the walls' rigidity, as they bend during compression and fail to hold the compliance beam in place, leading to monostable behavior. To solve this problem, a solution proposed during the design process is to create a wall-holder structure, bounding the walls toward the central structure of the component. The wall holder is created using the tangent function, in which one side is connected to the wall and the other side is connected to the center line of the component, with a radius of 42 mm. Through the experiment, the structure was proven to be effective in holding the walls in place during compression, increasing the rigidity of the walls.

There are three milestones during the design process that contribute to the success of fabricating the bistable component. The first milestone occurred at model 6, where the profile of the curve beam was changed from only using the 3-point arc method for the whole compliance beam to adding a hinge using the 3-point arc method and designing the curve beam using the tangent method to give more control over the angle, curvature, and thickness of the whole compliance beam. The second milestone occurred at model 7, when the side walls of the component were observed to have been deflected outward when under compression, preventing energy from the beam from being kept in a second stable state. Therefore, a change in the printing process was made to give the walls more rigidity, which is to increase the infill percentage from 20 to 50%. The result was observed to have achieved bistability. To improve the capability of reserving the energy of the compliance beam, an additional feature is added to the design at test 8 (the last milestone), which is the wall holder. It is designed as a curve connecting the walls to the base, which helps to reduce load on the walls when under compression and alter the outward force toward the base.

All the bistable systems have been modeled in the FEA ABAQUS software (Dassault Systems, Vélizy-Villacoublay,

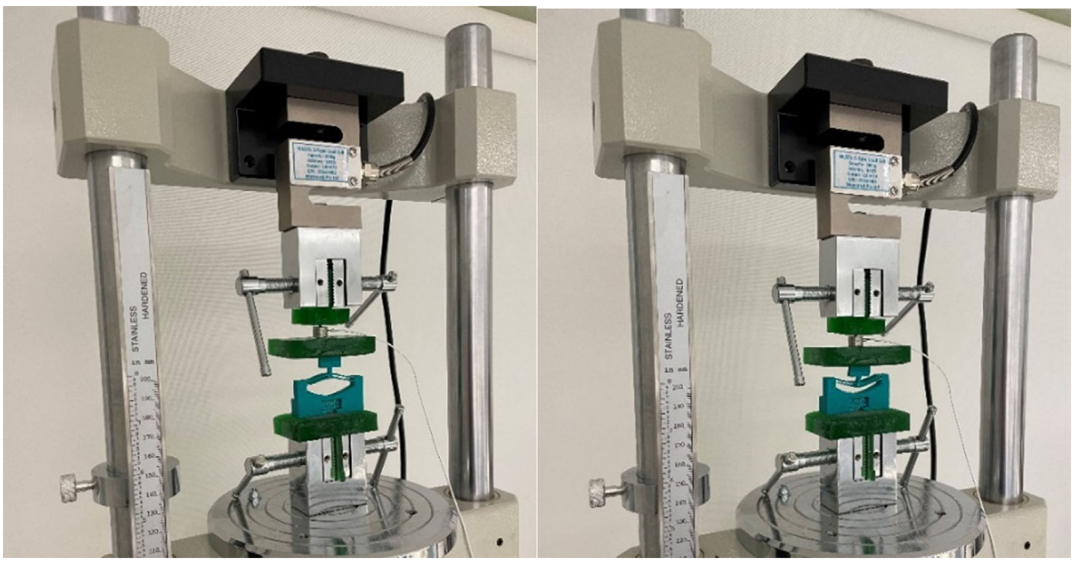

Figure 8. Force–displacement measurements of the 3D-printed bistable samples.

France) to find out the force displacement results. For simulation, the CAD of bistable parts is imported by STP files. According to the experimental test, the sample is compressed by an upper rigid plate at a velocity of $50\,\mathrm{mm\,min^{-1}}$. Also, the material properties of 3D-printed ABS were obtained with tensile tests according to the ASTM D638-14 standard with the elastic modulus of 2300 MPa, Poisson's ratio of 0.35, density of $1040\,\mathrm{kg\,m^{-3}}$, yield stress of 34.17 MPa, and strain for fracture of 7.7% with elastic and plastic behavior. As shown in **Figure 6**a, 4-node linear tetrahedron elements (C3D4) for the bistable part and 4-node 3-D bilinear rigid quadrilateral (R3D4) for the rigid plate were used in the present modeling. The number of elements is different according to the CAD. The size of the elements was chosen so that the results are independent of the mesh size.

From the graphs shown in **Figure 7**, it could be seen that designs 6, 8, and 9 were reflecting bistable characteristics where the load snapped through the negative values. However, as explained in Table 2, design 6 experiences near bistability (snap back to the first steady state shortly without external force applied in experimental tests). Both designs 8 and 9 satisfy bistability, but design 8 is shown to be fragile under fatigue in experimental tests, and the bottom beam breaks after several tests. The final design was also tested under compression loading

Table 3. Bistability test models.

Design #	Curve profile	Hinge angle	Hinge length	Print profile	Beam thickness	Results	Observations	Comments
1	3-point arc (38 mm radius)	75 deg from base	1.5 mm	120% scale, infill 20%	1.5 mm	Monostable; beam is flexible; no fracture was found	High damping effect (require larger force for linear motion)	Missing connection block between upper and bottom beam
2	3-point arc (38 mm radius)	85 deg from base	1.5 mm	120% scale, infill 20%	1.5 mm	Monostable	Break at peak point of the curve	Missing connection block between upper and bottom beam; monostable can still be observed for the initial stress tests; however, the beam breaks after several tests
3	3-point arc (38 mm radius)	90 deg from base	1.5 mm	120% scale, infill 20%	1.5 mm	Monostable; beam is flexible; no fracture was found	Nil	Hinge is not curve/long enough to hold the curve beam at the second stable state
4	3-point arc (38 mm radius)	90 deg from base	1.5 mm	100% scale	1 mm	Broken hinge during compression	Nil	Hinge breaks during compression
5	Tangent and 3-point arc (11 mm)	Tangent angle (26 deg)	Tangent curve length (20 mm)	120% scale, infill 20%	2 mm	Beam is not flexible and does not respond to input force	Nil	Thick beam prevents the ability to bend the structure, curvature of the beam, and hinges preventing the beam from bending
6	Tangent and 3-point arc (23 mm)	Tangent angle (15 deg)	Tangent curve length (20.8 mm)	120% scale, infill 20%	0.8 mm	Monostable, near bistability (hold at second steady state for 30 s and then snap back to first steady state without external force applied	High damping effect (require larger force for linear motion)	Two steady states provide a 5 mm stroke
7	Tangent and 3-point arc (23 mm)	Tangent angle (12 deg)	Tangent curve length (20.8 mm)	120% scale, infill 20%	0.8 mm	Broken beam	High damping effect (require larger force for linear motion)	Hinge too short causing more curvature to the beam, leading to more tension applied at peak point when bend, causing the beam to break
8	Tangent, 3-point arc (23 mm), and wall holder (3-point arc 42 mm)	Tangent angle (15 deg)	Tangent curve length (20.8 mm)	120% scale, infill 50%	0.8 mm	Bistability achieved; bottom beam breaks after several tests	High damping effect (require larger force for linear motion)	Beam thickness might be too much causing fracture during compression
9	Tangent, 3-point arc (23 mm), wall holder (3-point arc 42 mm), and widen connecting beam from 0.5 mm to 2 mm	Tangent angle (15 deg)	Tangent curve length (20.8 mm)	120% scale, infill 50%	0.6 mm	Bistability achieved; bottom beam is flexible; no fracture was found	High damping effect (require larger force for linear motion)	Final design for bistable component

experimentally to measure the maximum force before snapping through (**Figure 8** and **Table 3**).

A vibration test is then conducted to assess the performance of the bistable component in changing the working frequency of the TMD system. The experiment is set up by attaching a flexible stainless steel ruler to a shaker (TIRA TV 5220/LS-12, Germany) at one end as a reference point where the external force would be applied to vibrate the system (**Figure 9**). Two accelerometer sensors are placed at the reference point and the main mass to observe their acceleration and displacement according to the frequency of the shaker. The TMD mass is attached to the ruler on the other end, and connecting to it is a stack of five bistable components to provide the stroke needed for the TMD shift to the desired location, which gives the total stroke length of about 3.3 cm. Each unit is firmly attached to the ruler by a housing at the back to hold it in place. Figure 9b demonstrates an application of the proposed TMD method for satellite panel vibration isolation.

The initial test is conducted with only the $m_{\text{main}} = 150$ g to verify the natural frequency of the system by varying the induced frequency of the shaker from 5 to 50 Hz. The peak point of the system in terms of acceleration and displacement in the frequency domain graph generated from the test can be seen in **Figure 10** and **11**, respectively. The next test is conducted with the TMD mass system attached to the ruler on the other end with the bistable component in its compressed mode and another test in its extended mode. These are to define the working frequency of the TMD system as well as to demonstrate how the TMD system can effectively reduce the amplitude of the acceleration over a range of frequencies.

Once the most optimized bistable component has been designed and fabricated, it is implemented in the TMD mass of the vibration test to help shift the TMD mass position,

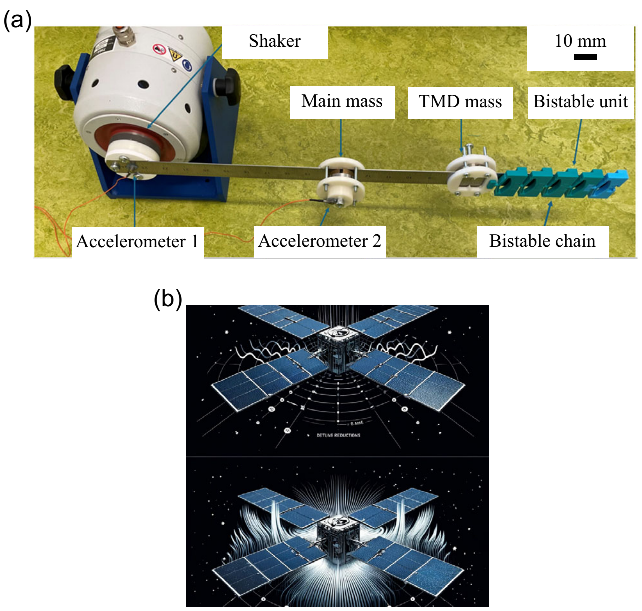

Figure 9. a) Vibration test set up for assessing the effect of the proposed enhanced TMD with stacked bistable cells. b) Satellite panel vibration control of the bistable TMD system.

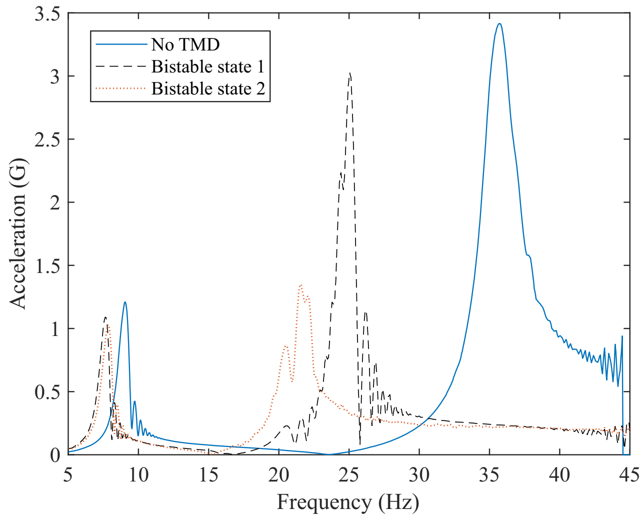

Figure 10. Effect of TMD bistable mechanism on acceleration of the system.

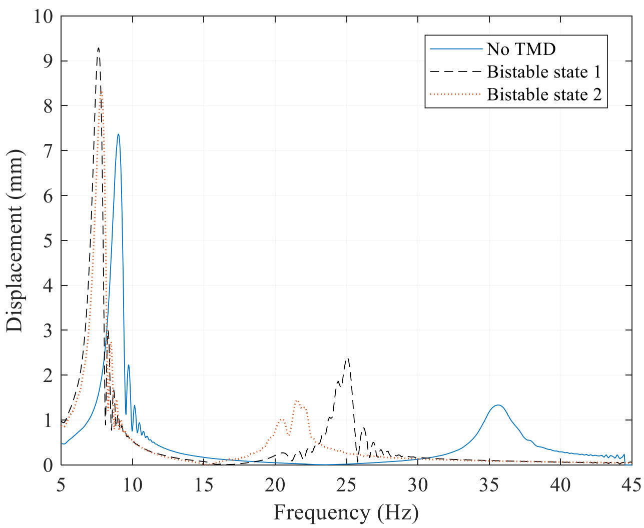

Figure 11. Effect of TMD bistable mechanism on displacement of the system.

resulting in the improvement of the vibration mitigation effect as shown in Figure 10 and 11.

During the first test, only the main mass of 150 g is attached to the system. With the induced vibration from the other end of the structure, the main mass suffers from the vibration intensively during the system's natural frequency of about 35 Hz. As shown, the acceleration force measured from the main mass reaches about 3.5G. The second test is conducted with the TMD mass introduced and connected to the bistable component in its compressed mode, bringing the TMD mass outside of the working frequency of the system. As a result, the natural frequency of the system is reduced to around 25 Hz, with its amplitude slightly dropping to 3G. The last test is conducted with the same TMD mass, but the bistable component is now extended, bringing the TMD mass closer to the main mass and inside the

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working range of the system's natural frequency. The results show that, while the system's frequency peak point is shifted down to 20 Hz, its acceleration peak point also decreases to only 1.3G, which has 63% lower effects when under the same range of oscillation frequency.

The results presented in this study demonstrate the significant potential for performance tuning enabled by the integration of bistable mechanisms in TMD systems. Through adaptive repositioning of the TMD mass, the system can dynamically adjust its natural frequency by modifying the local stiffness characteristics of the flexible beam. This passive adjustment mechanism offers a unique advantage over conventional TMD systems, which are typically optimized for a fixed frequency range. Furthermore, the ability to transition between bistable states allows the system to maintain vibration suppression effectiveness across a broader range of operational conditions. This feature is particularly beneficial in scenarios where the vibrational frequency profile changes over time or under varying external loads. While the present study focuses on demonstrating the effectiveness of this adaptive mass tuning approach, future investigations will explore the integration of smart materials and real-time feedback systems to further refine and expand the system's performance tuning capabilities. This opens avenues for the development of next-generation vibration control systems capable of self-adjusting to dynamic environments while maintaining structural efficiency and reliability.

4. Conclusion

The integration of modular bistable mechanisms in vibration isolation systems through 3D printing offers a significant leap in design flexibility and functionality. This work has been devoted to optimizing the design of the compliance beam. By carefully selecting the material's stiffness and refining geometric configurations, the stroke length has been effectively tailored to meet the specific requirements of the TMD vibration test, offering precise and reliable performance. By utilizing 3D printing techniques, complex bistable structures, capable of switching between two stable states under specific stimuli (external force here) can be precisely fabricated. This capability enhances vibration isolation by allowing dynamic adjustment of system parameters in response to varying vibrational frequencies, thereby broadening the effective operational bandwidth of such systems. The combination of bistable designs with additive manufacturing techniques paves the way for innovative, adaptive vibration control solutions. Mass positioning and stiffness adjustment were prioritized as the primary contributors to vibration control in this work, given their direct influence on frequency alignment and resonance tuning within the TMD system. While damping effects play a critical role in overall performance, they could be future work directions to quantify their contribution and further optimize the system's vibration suppression capabilities.

Future research could explore the integration of smart materials into the bistable component via 4D printing, thereby having more control over the stroke lengths between different stable states of the compliance beam and enabling a self-triggering mechanism between stable states. This improvement holds immense potential for further enhancing the efficacy and adaptability of mechanical vibration control systems, paving the way for more robust and responsive structural dynamics management in diverse engineering applications.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Ali Zolfagharian: conceptualization (lead); data curation (lead); formal analysis (lead); funding acquisition (lead); investigation (lead); methodology (lead); project administration (lead); resources (lead); supervision (lead); writing—original draft (lead); writing—review & editing (lead). **Frédéric Demoly**: formal analysis (equal); validation (equal); writing—review & editing (equal). **Mohammad Lakhi**: data curation (equal); software (equal); visualization (equal); writing—original draft (supporting). **Bernard Rolfe**: supervision (supporting). **Mahdi Bodaghi**: supervision (supporting); validation (equal); writing—review & editing (supporting).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

4D printing, bistable, compliant, tuned mass dampers, vibrations

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[1] Y.-S. Chang, V. N. D. Kieu, S.-C. Huang, *Micromachines* **2022**, *13*, 817.
[2] K. Alsaif, M. A. Foda, *J. Sound Vib.* **2002**, *256*, 629.
[3] P. Le, D. Xu, A. Vazhapilli Sureshbabu, M. Zimmermann, *Proc. Des. Soc.* **2023**, *3*, 3801.
[4] M. Rezaei, R. Talebitooti, W.-H. Liao, M. I. Friswell, *J. Sound Vib.* **2023**, *546*, 117413.
[5] J. Zhang, Y. Zhu, J. Tu, Z. Li, Q. Wang, *Materials* **2022**, *15*, 1829.
[6] A. R. Damanpack, M. Bodaghi, W. H. Liao, *Int. J. Non-Linear Mech.* **2019**, *108*, 87.
[7] F. Liu, X. Jiang, X. Wang, L. Wang, *Extreme Mech. Lett.* **2020**, *41*, 101002.
[8] B. Hu, X. Fang, J. Wen, D. Yu, *Int. J. Mech. Sci.* **2024**, *272*, 109172.
[9] K. Wu, *Design, Modeling, Optimization and Control of Compliant Mechanisms*, Centrale Lille, Lille **2023**.

ADVANCED ENGINEERING MATERIALS
www.aem-journal.com

[10] M. Koh, Y. Wang, K. Shea, T. Chen, *Eng. Struct.* **2023**, *295*, 116792.
[11] M. Liu, W. Wang, Y. Zhu, Y. Yuan, Y. Niu, L. Dong, C. Wang, K. Jiang, G. Chen, *Sci. China Technol. Sci.* **2023**, *66*, 3186.
[12] A. Mukherjee, S. Sudersan, S. F. Ali, A. A, *Smart Mater. Struct.* **2021**, *30*, 075025.
[13] L. Wu, D. Pasini, *Adv. Mater.* **2023**, *35*, 2301109.
[14] C. Perez-Garcia, J. Aranda-Ruiz, M. L. Lopez-Donaire, R. Zaera, D. Garcia-Gonzalez, *Adv. Funct. Mater.* **2024**, 2313865.
[15] M. A. Ten Wolde, D. Farhadi, *Mech. Mach. Theory* **2024**, *196*, 105626.
[16] S. A. Zirbel, K. A. Tolman, B. P. Trease, L. L. Howell, M. S. Kellermayer, *PLoS One* **2016**, *11*, e0168218.
[17] S. Dalela, P. S. Balaji, D. P. Jena, *Nonlinear Dyn.* **2022**, *108*, 1931.
[18] A. Zolfagharian, P. Picken, M. Bodaghi, M. Fard, B. Rolfe, *Adv. Eng. Mater.* **2023**, *25*, 2300356.
[19] H. Fan, L. Yang, Y. Tian, Z. Wang, *Compos. Struct.* **2020**, *243*, 112244.
[20] H. Y. Jeong, S.-C. An, Y. Lim, M. J. Jeong, N. Kim, Y. C. Jun, *Appl. Sci.* **2020**, *10*, 7254.
[21] H. Y. Jeong, S.-C. An, I. C. Seo, E. Lee, S. Ha, N. Kim, Y. C. Jun, *Sci. Rep.* **2019**, *9*, 324.
[22] G. Risso, M. Kudisch, P. Ermanni, C. Daraio, *Adv. Sci.* **2024**, *11*, 2308903.
[23] G. Scalet, *Actuators* **2020**, *9*, 10.
[24] K. Liang, Y. Wang, Y. Luo, A. Takezawa, X. Zhang, Z. Kang, *Mater. Des.* **2023**, *227*, 111810.
[25] Y.-Y. Chen, Z.-C. Qian, W. Zhao, C.-M. Chang, *J. Sound Vib.* **2020**, *473*, 115233.
[26] M. Li, D. Yu, Y. Li, X. Liu, F. Dai, *Compos. Struct.* **2023**, *314*, 116971.
[27] J. Qiu, J. H. Lang, A. H. Slocum, *J. Microelectromech. Syst.* **2004**, *13*, 137.
[28] H. Yang, L. Ma, *J. Mater. Sci.* **2019**, *54*, 3509.

